

Announcement

Kindly note: We're currently experiencing a high volume of inquiries and calls, potentially leading to delays in our initial response.



Dermatology 23S Module 2

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Day 2-3 Scenario 2

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[◀ Follow on Questions](#)[Scenario 3 ▶](#) Display replies in nested form**Day 2-3 Scenario 2**

by [Derek Kayanja \(Tutor\)](#) - Sunday, 12 November 2023, 3:09 PM

Hi group,

Here is our second scenario

A 68-year-old female attends clinic complaining of a tender nodule on the right pinna for 3 months. The lesion is tender and scaly and there is a small amount of bleeding on the pillow.

In her past medical history she had a right-sided hemiparesis due to a CVA 3 years ago. She is receiving losartan, aspirin and simvastatin.

On examination, she has a 7 × 8 mm scaly nodule on the left upper pole of the pinna with a superficial area of scaling.



Please answer all the questions below:

1. What bedside test could be performed to assist the diagnosis?
2. What is the likely diagnosis?

3. What is the most appropriate treatment for this patient?

120 words

Reply



Re: Day 2-3 Scenario 2

by [Adam Tollitt](#) - Sunday, 12 November 2023, 5:51 PM

Chondrodermatitis nodularis helicis

The most likely diagnosis of this lesion is chondrodermatitis nodularis helicis.

These are benign, painful, solitary, small nodules found on either the ear helix or antihelix. They typically have a scaly/crusted surface with an underlying area of ulceration; with the nodule surrounded by an area of erythema (Eshraghi and Oakley, 2018; Cunliffe, 2022).

I suspect that the patient having a right hemiparesis from her previous stroke may mean that she is prone to sleeping with the left side of her head against her pillow and that this may have contributed to the development of the chondrodermatitis nodularis helicis.

Dermatoscopy would be a useful aid in the diagnosis of this lesion, with the characteristic feature of this lesion on dermatoscopy being a 'daisy pattern' (García-García et al., 2018).



Figure 1 – Dermatoscopic image showing the characteristic 'daisy pattern' of white radial thick lines surrounding a central rounded area of crust/erosion (García-García et al., 2018).

Treatment options for chondrodermatitis nodularis helicis can be divided into conservative, medical and surgical options as below (Eshraghi and Oakley, 2018; Cunliffe, 2022).

Conservative:

- Sleeping on the non-affected side

- Foam pillows/pillows with holes in to prevent the ear being pressed against the pillow
- Wearing warm hats outdoors

Medical:

- Intralesional steroid injection – reduces localised inflammation
- Topical nitroglycerin – causes local arteriolar vasodilation, resulting in less ischaemic burden on the lesion
- Cryotherapy

Surgery:

- Curettage
- Surgical excision
- Punch biopsy excision

For this particular patient it may be appropriate to trial initially one of the conservative management options; with a foam or specially adapted pillow perhaps the most appropriate initial conservative management option.

Question for the group

Can anyone expand on the pathophysiology of chondrodermatitis nodularis helices and some of the associated medical conditions with this lesion?

References

Cunliffe, T. (2022). *Chondrodermatitis nodularis helices*. [online] Primary Care Dermatology Society. Available at: <https://www.pcds.org.uk/clinical-guidance/chondrodermatitis-nodularis-helices>. (accessed 12/11/2023).

Eshraghi, A. and Oakley, A. (2018). *Chondrodermatitis nodularis helices (CNH) | DermNet NZ*. [online] Available at: <https://dermnetnz.org/topics/chondrodermatitis-nodularis>. (accessed 12/11/2023).

García-García, B., Munguía-Calzada, P., Aubán-Pariente, J., Junceda-Antuña, S., Zaballos, P., Argenziano, G. and Vázquez-López, F. (2018). Dermoscopy of chondrodermatitis nodularis helices. *Archives of Dermatological Research*, [online] 310(7), pp.551–560. doi:<https://doi.org/10.1007/s00403-018-1844-6>. (accessed 12/11/2023).

355 words

Reply



Re: Day 2-3 Scenario 2

by [Zarmeena Khan](#) - Monday, 13 November 2023, 8:43 AM

Hi Adam, excellent discussion and spot diagnosis,
I agree with the diagnosis of Chondrodermatitis nodularis helices (CNH) in my clinical practice, I feel it's a relatively uncommon condition, as you have well explained it's a painful, inflammatory condition that affects the skin and cartilage of the ear, particularly the helix.

As for your question pathophysiology is not fully understood, but it's believed to be related to pressure, friction, and reduced blood supply to the skin and cartilage of the ear. Continuous pressure and trauma (such as from sleeping on one side) may lead to the development of micro-injuries and subsequent inflammation in the area. Additionally, factors like sun exposure, cold temperatures, and microtrauma are thought to contribute to the development of CNH.

While reading on the topic, i came across **Perichondrial vasculitis theory**, described in 2009 by Upile et al., which proposes that the disease process begins as a result of pressure on the auricle, potentially exacerbated by same-side sleeping and prominauris, which subsequently causes arteriolar narrowing in the perichondrium of the helix. This leads to ischemia, necrosis, and extrusion of underlying cartilage. Ultimately, a foreign body reaction ensues, resulting in a severe localized inflammatory reaction and the development of CNH. (Gupta, Hohman and Kwan, 2023)

Several medical conditions have been associated with chondrodermatitis nodularis helcis, including:

1. Connective Tissue Disorders: Conditions like lupus erythematosus and rheumatoid arthritis have been observed alongside CNH. Connective tissue disorders might contribute to the development of CNH.
2. Vascular Insufficiency: Any condition that affects blood supply or circulation could potentially exacerbate or contribute to CNH due to the role of reduced blood flow in its pathogenesis.
3. Environmental Factors: Prolonged exposure to cold, sun damage, and pressure on the ear (commonly from sleeping on one side) are environmental factors that might trigger or worsen CNH.
4. Genetic Predisposition: While not fully elucidated, genetic factors might contribute to an individual's susceptibility to CNH.

References:

1. Goldberg, L. H., & O'Hare, A. (2005). Chondrodermatitis Nodularis Chronica Helicis. eMedicine. <https://emedicine.medscape.com/article/1054255-overview#a3>
2. Wilkin, G., Greer, K., & Bernstein, M. L. (1994). Chondrodermatitis Nodularis Helicis as a Sign of Underlying Connective Tissue Disease. JAMA Dermatology, 130(6), 747-750. <https://doi.org/10.1001/archderm.1994.01690060087018>
3. Gupta, G., Hohman, M.H. and Kwan, E. (2023). *Chondrodermatitis Nodularis Helicis*. [online] Nih.gov. Available at: <https://www.ncbi.nlm.nih.gov/books/NBK482507/> [Accessed 13 Nov. 2023].

381 words

Reply



Re: Day 2-3 Scenario 2

by [Adam Tollitt](#) - Monday, 13 November 2023, 9:32 AM

Hi Zarmeena,

Thank you for answering my question about CNH pathophysiology and related conditions.

Your discussion about the 'perichondrial vasculitis theory' was very interesting and does help to provide a possible explanation for why CNH develops secondary to recurrent trauma/pressure from same side sleeping.

Question for the group:

In my reading I noted that CNH appears to affect middle-age to elderly patients; with males being more commonly affected than females (Eshraghi and Oakley, 2018; Salah, Urso and Khachemoun, 2018).

Is anyone able to discuss/hypothesise why CNH affects predominantly 1) middle-age to elderly patients and 2) males more than females?

Addendum:

Apologies, in my initial post I have noticed an error in my explanation for why the patient developed the CNH lesion on her right pinna. The explanation should read 'prone to sleeping with the right side of her head against the pillow' (due to right hemiparesis).

References

Eshraghi, A. and Oakley, A. (2018). *Chondrodermatitis nodularis helices (CNH) | DermNet NZ*. [online] Available at: <https://dermnetnz.org/topics/chondrodermatitis-nodularis>. (accessed 13/11/2023).

Salah, H., Urso, B. and Khachemoune, A. (2018). Review of the Etiopathogenesis and Management Options of Chondrodermatitis Nodularis Chronica Helicis. *Cureus*, [online] 10(3). doi:<https://doi.org/10.7759/cureus.2367>. (accessed 13/11/2023).

192 words

Reply



Re: Day 2-3 Scenario 2

by [Pooran Outar](#) - Monday, 13 November 2023, 4:29 PM

Hi Adam,

Excellent discussion on CNH.

With regards to your question, I would like to hypothesize the following:

Middle-aged and elderly men are more susceptible to CNH due to anatomical changes in the ear, increased sun exposure, and alterations in the microscopic structure of ear cartilage. The prevalence of CNH in this demographic group, ranging from 70 to 90% (Jordan, 2023), may be attributed to the combination of larger and more protruding ears in males, along with the observed age-related increase in ear size (Japatti et al., 2018). The anatomical features, such as a protruding outer ear rim, may contribute to higher sun exposure, potentially increasing the risk for CNH. Furthermore, age-related changes in the microscopic structure of ear cartilage, characterized by decreased elastic fibres, vascular supply, cartilage cell density, skin elasticity, and resilience, could render the cartilage more susceptible to trauma or pressure-related injuries (Japatti et al., 2018). These structural changes may impede adequate healing and contribute to the development of secondary perichondritis (Marks, 2021), ultimately leading to the higher prevalence of CNH in middle-aged and elderly men.

References:

- Japatti, S.R., Engineer, P.J., Reddy, B.M., Tiwari, A.U., Siddegowda, C.Y. and Hammannavar, R.B. (2018). Anthropometric Assessment of the Normal Adult Human Ear. *Annals of Maxillofacial Surgery*, [online] 8(1), pp.42–50. doi:https://doi.org/10.4103/ams.ams_183_17.
- Jordan, C. (2023). *What Is Chondrodermatitis?* [online] WebMD. Available at: <https://www.webmd.com/skin-problems-and-treatments/what-is-chondrodermatitis> [Accessed 13 Nov. 2023].
- Marks, V.J. (2021). Chondrodermatitis Nodularis Helicis: Background, Pathophysiology, Etiology. *eMedicine*. [online] Available at: <https://emedicine.medscape.com/article/1119141-overview#a5> [Accessed 13 Nov. 2023].

244 words

Reply



Re: Day 2-3 Scenario 2

by [Xiao Liey Ding](#) - Tuesday, 14 November 2023, 9:31 AM

Hi Adam,

There is not a comprehensive understanding of why Chondrodermatitis Nodularis Helicis (CNH) predominantly affects middle-aged to elderly patients and shows a higher incidence in males compared to females. The literature on CNH etiology is somewhat limited, and specific factors contributing to age and gender distribution remain speculative.

Here are some etiopathogenesis by Salah, H. et al (2018) for the reference and to further extrapolate why middle-aged to elderly patients are more prone to CNH.

1. Cumulative Exposure: One hypothesis is that CNH develops over time due to cumulative exposure to chronic pressure on the ear. Middle-aged and elderly individuals may have had more prolonged exposure to factors such as sleeping on one side, using earphones, or wearing tight headgear.
2. Skin Changes with Aging: Aging is associated with changes in skin elasticity and thickness. The skin on the ear may become less resilient with age, potentially making it more susceptible to the development of CNH.
3. Reduced Cartilage Resilience: Age-related changes in cartilage elasticity might contribute to the formation of nodules. The ear cartilage could become less resilient, making it more prone to damage.

As of why male are affected more than female, any takers?

Reference:

Salah, H., Urso, B., and Khachemoune, A. (2018). Review of the Etiopathogenesis and Management Options of Chondrodermatitis Nodularis Chronica Helicis. *Cureus*, 10(3), e2367. Doi: [10.7759/cureus.2367](https://doi.org/10.7759/cureus.2367)

223 words

Reply



Re: Day 2-3 Scenario 2

by [Zarmeena Khan](#) - Tuesday, 14 November 2023, 7:24 PM

Hi Xiao,

Very interesting discussion,

i looked up on the evidence for prevalence of CNH in males vs female populations,

The information about chondrodermatitis nodularis (CNH) being more prevalent in males is not well-established or strongly supported by extensive scientific literature. In fact, there is limited epidemiological data specifically addressing gender prevalence in CNH.

Some literature does mention that CNH seems to be more common in males, but the evidence supporting this observation is limited. Possible reasons for a perceived male predominance might include:

1. Occupational Factors:

- Some studies suggest that occupational factors, such as prolonged pressure on the ear during sleep or work-related activities, may contribute to the development of CNH. If certain occupations or activities with increased ear pressure are more common among males, it could influence the observed gender distribution.

2. Hormonal Factors:

- Hormonal influences may play a role, as there can be differences in skin structure and response to trauma between males and females. However, more research is needed to establish a clear connection between hormones and CNH.

3.

Risk Factors and Behavior:

-Some studies suggest that pressure and trauma to the ear, which are considered potential contributors to the development of CNH, might be more common in males. For instance, sleeping on the affected side or prolonged pressure on the ear due to habits could be more prevalent in males, contributing to the development of CNH.

- References:

1.Jiam, N. T., & Goldberg, L. H. (2014). Chondrodermatitis Nodularis Helicis: A Review. Dermatologic Surgery, 40(11), 1163–1168. DOI: [10.1097/dss.000000000000180] (<https://doi.org/10.1097/dss.000000000000180>)

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3.Goldstein, B. G., & Goldstein, A. O. (2012). Chondrodermatitis Nodularis Helicis in an Adolescent Female. Pediatric Dermatology, 29(4), 495–496. DOI: [10.1111/j.1525-1470.2011.01465.x](https://doi.org/10.1111/j.1525-1470.2011.01465.x)

4.Tsai, T. F., & Kuo, T. (1993). Chondrodermatitis nodularis chronica helicis. A study of 164 Chinese patients. Dermatologica Sinica, 11(1), 7–13.

323 words

Reply

**Re: Day 2-3 Scenario 2**

by [Derek Kyanja \(Tutor\)](#) - Wednesday, 15 November 2023, 12:00 PM

Xiao Liey, thank you very good response.

7 words

Reply

**Re: Day 2-3 Scenario 2**

by [Sadaf Basharat](#) - Sunday, 19 November 2023, 5:49 PM

Hi Adam.

First of all I would like to say that you have actually highlighted a very interesting fact regarding CNH and upon searching for this i have found that usually the hearing issues are observed in the middle aged and elderly populations, for which they usually use the hearing aid, which is a probable cause of CNH. Along with that the anatomical and skin changes related to ageing also make CNH to be observed more commonly in middle aged population. Sleeping habits of sleeping to one side also makes an individual, more susceptible of developing CNH, however the association of CNH predominantly with men has been observed in multiple studies but unfortunately i could not go through any actual reasoning for it.

I have a question for my peers that has anyone ever witnessed a CNH in a child?

References:

1. Salah, H., Urso, B. and Khachemoune, A. (2018). Review of the Etiopathogenesis and Management Options of Chondrodermatitis Nodularis Chronica Helicis. Cureus, [online] 10(3). doi:<https://doi.org/10.7759/cureus.2367>. (accessed 13/11/2023).

2. NEWCOMER, V.D., STEFFEN, C.G., STERNBERG, T.H. and LICHTENSTEIN, L., 1953. Chondrodermatitis nodularis chronica helicis: Report of ninety-four cases and survey of literature, with emphasis upon pathogenesis and treatment. AMA Archives of Dermatology and Syphilology, 68(3), pp.241-255.

205 words

Reply

**Re: Day 2-3 Scenario 2**by [Derek Kayanja \(Tutor\)](#) - Wednesday, 15 November 2023, 11:59 AM

Thank you Zarmeena very good addition to our discussion

9 words

[Reply](#)
**Re: Day 2-3 Scenario 2**by [Derek Kayanja \(Tutor\)](#) - Wednesday, 15 November 2023, 11:58 AM

Thank you Adam, very good post and yes this is chondrodermatitis nodularis helioides.

13 words

[Reply](#)
**Re: Day 2-3 Scenario 2**by [Sandeep Kumar](#) - Wednesday, 15 November 2023, 2:13 PM

Hi Adam, Good work there.

I was looking for pathophysiology of CDNH and found this interesting article mentioning the causes of this condition. Most likely the pathophysiology is thru- Perforating dermatoses and transepidermal elimination. Other causes which play major role are-

Age, Cartilage degeneration, Ear anatomy, Genetics, Glomus-like vascular changes, Pressure etc.

Although this is idiopathic- can be associated with Autoimmune/ connective disorders especially in pediatric and female groups.

A study found 88.2% of patients with CDNH with raised cholesterol and lipid levels. So cardiovascular health is also associated with it.

References-

Salah H, Urso B, Khachemoune A (March 26, 2018) Review of the Etiopathogenesis and Management Options of Chondrodermatitis Nodularis Chronica Helicis. Cureus 10(3): e2367. doi:10.7759/cureus.2367

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Karabulut YY, Senel E, Dölek Y: Frequency and etiology of chondrodermatitis nodularis chronica helicis. Indian J Otol. 2013, 19:140-142. 10.4103/0971-7749.117471

Sehgal VN, Singh N: Chondrodermatitis nodularis. Am J Otolaryngol. 2009, 30:331-336. 10.1016/j.amjoto.2008.04.001

190 words

[Reply](#)
**Re: Day 2-3 Scenario 2**by [Siddharth Saluja](#) - Saturday, 18 November 2023, 11:15 AM

Hi Adam,

Most certainly.

We know that Chondrodermatitis nodularis helicis (CNH) is a condition that primarily affects the cartilage of the ear. The exact pathophysiology of CNH is not fully understood, but it is thought to be multifactorial in nature.(1)

One proposed theory suggests that CNH develops due to a combination of mechanical trauma, ischemia (reduced blood supply), and inflammation. Continuous pressure or friction on the cartilage, often from

sleeping on the affected side, may cause microtrauma and disrupt the blood flow to the region. This compromised blood supply can subsequently lead to tissue damage and inflammation, resulting in the formation of a painful nodule or lesion. (1)

Certain medical conditions have been associated with CNH, although the direct cause-and-effect relationship is not always clear. These conditions include 🧐 1)

Actinic damage: Chronic exposure to sunlight and ultraviolet radiation can contribute to the development of CNH lesions, especially in individuals with fair skin or a history of excessive sun exposure.

Connective tissue disorders: Some studies have reported an association between CNH and connective tissue disorders like rheumatoid arthritis, systemic lupus erythematosus, or Ehlers-Danlos syndrome. The exact mechanism linking these conditions to CNH is still being investigated.

Immunocompromised states: Immune system dysfunction, such as in HIV/AIDS or immunosuppressive therapy, may increase the risk of developing CNH. It is believed that the compromised immune response could lead to impaired tissue healing or increased susceptibility to trauma.

Microvascular pathology: Conditions that affect blood vessels, such as diabetes or vasculitis, have been linked to CNH. These vascular abnormalities could contribute to decreased blood flow to the cartilage, further exacerbating the development of CNH lesions.

It's important to note that while these conditions have been associated with CNH, they are not always present in every case. CNH can also occur in individuals without any underlying medical conditions. (2)

References:

1 Gupta, G. and Badri, T. (2020). Chondrodermatitis Nodularis Helicis. [online] PubMed. Available at: <https://www.ncbi.nlm.nih.gov/books/NBK482507/>.

2)dermnetnz.org. (n.d.). Chondrodermatitis nodularis helicis (CNH) | DermNet NZ. [online] Available at: <https://dermnetnz.org/topics/chondrodermatitis-nodularis>.

329 words

Reply



Re: Day 2-3 Scenario 2

by [Pooran Outar](#) - Monday, 13 November 2023, 2:56 PM

Diagnosis: Chondrodermatitis Nodularis Helicis

The bed side test that can aid in diagnosis is the **"pressure test"**, since applying pressure to the suspected lesion may reproduce pain (The British Association of Dermatologists, 2023).

The treatment for CNH involves a combination of general measures, medical treatments, and, in some cases, surgical options (Eshraghi and Oakley, 2018).

General measures include:

- Using protective padding at night
- Avoiding sleeping on one side
- Wearing a soft pillow or a specially crafted ear protector.
- Protecting the ears in cold weather
- Applying petroleum jelly or antiseptic ointment

Medical treatments include:

- Intralesional steroid injections to reduce inflammation
- Implanting fillers for cushioning
- Using topical nitroglycerin ointment to improve blood flow.

Surgical options range from:

- Removing skin and cartilage
- Excision with skin grafting
- Curettage

Based on a study by Juul Nielsen, Holkmann Olsen and Lock-Andersen, 2016, I would suggest General measures and Intralesional steroids as first treatment options for this patient and evaluate after 8 weeks, then consider surgical options if no success.

I would appreciate any feedback on the thought process I outlined above for managing this case.

References:

- Eshraghi, A. and Oakley, A. (2018). *Chondrodermatitis nodularis helcis (CNH) | DermNet NZ*. [online] dermnetnz.org. Available at: <https://dermnetnz.org/topics/chondrodermatitis-nodularis>.
- Juul Nielsen, L., Holkmann Olsen, C. and Lock-Andersen, J. (2016). *Therapeutic Options of Chondrodermatitis Nodularis Helcis*. [online] Plastic Surgery International. Available at: <https://www.hindawi.com/journals/psi/2016/4340168/>.
- The British Association of Dermatologists (2023). *British Association of Dermatologists*. [online] Bad.org.uk. Available at: <https://www.bad.org.uk/pils/chondrodermatitis-nodularis/> [Accessed 13 Nov. 2023].

240 words

Reply



Re: Day 2-3 Scenario 2

by [Derek Kayanja \(Tutor\)](#) - Wednesday, 15 November 2023, 12:01 PM

Thank you Pooran, good post and correct diagnosis.

8 words

Reply



Re: Day 2-3 Scenario 2

by [Rahima Abdulaziz Ahmed](#) - Monday, 13 November 2023, 4:56 PM

Thank you Adam for the dermoscopy photo of the condition , and Zarmeena for the addition.

Q1;

The bedside test I could perform to assist in the diagnosis involves applying pressure to the the painful nodule. If the nodule is tender and reproduces the patient pain, it supports the diagnosis of CNH.

Q2;

Chondrodermatitis nodularis helcis also known as Winkler disease. (Gupta, Hohman and Kwan. 2023)

Its a benign inflammatory condition that affects the skin and cartilage of the pinna. The exact etiology is unknown but it is believed to be caused by chronic and excessive pressure on the pinna. This can include consistently sleeping on the same side, continuous and prolonged use of hearing aids, headphones, and other headgear. May also be associated with other conditions as discussed by Zarmeena.

Signs and symptoms include; spontaneously appearing painful nodule on the helix or antihelix of the pinna. Classically appears unilaterally but bilateral involvement has also been observed. Primarily occurs on the ear of the usual sleeping side, more often the right one. Common site is the apex of the helix , antihelix more common in women.

On Examination ; reveals well circumscribed , rounded nodule with raised edges and a crust in its center, often underlying an ulcer which may occasionally have exposed cartilage at the bottom. Lesion has a diameter of 4-6mm and is typically surrounded by an erythematous area.

Q3; Treatment for this patient;

Given her age and the comorbid conditions currently will go for conservative treatment for her for a month and see results , which includes;

1.Pressure relieving prosthesis and padding , the primary goal for these is to relieve or eliminate pressure on the lesion. They are available in various forms, such as self-adherent foam sponges, foam bandages attached to the head, and doughnut pillows for sleeping. These methods are met with variable success, predominantly dependent upon patient compliance rates.

if no improvement , may consider other options as discussed earlier by Adam.

My question to the group; what are the complications and differential diagnoses of chondrodermatitis nodularis helcis?

References;

Gupta, G. Hohman, H and Kwan, E. (2023.) *Chondrodermatitis Nodularis Helicis* StatPearls [Internet]. Available at <https://www.ncbi.nlm.nih.gov/books/NBK482507/> (Accessed on 13/11/23)

Marks, J (2020) *Chondrodermatitis Nodularis Helicis* Medscape (online) Available at <https://emedicine.medscape.com/article/1119141-overview#a2> (Accessed on 13/11/23)

376 words

Reply



Re: Day 2-3 Scenario 2

by [Oyeombiliyetu Jason](#) - Monday, 13 November 2023, 9:32 PM

Thank you for the thorough explanation I'd like to answer the question posed: Complications of Chondrodermatitis nodularis helicis (CNH) may include chronic discomfort leading to insomnia, superimposed infections, and potential complications related to treatment. Surgical interventions can carry the risk of non-healing lesions and the development of new CNH lesions.

Differential diagnosis for CNH involves considering conditions such as basal cell carcinoma (characterized by a daisy pattern on dermatoscopy), cutaneous squamous cell carcinoma (requires biopsy for differentiation), gouty tophi (multiple lesions on various sites), keratoacanthoma (rapid growth and spontaneous resolution), seborrheic or keratotic dermatitis (multiple lesions with diverse morphology), atypical fibroxanthoma, elastotic nodules of the ear, perforating dermatosis, pseudocyst of the auricle, cystic chondromalacia, and warts (skin-colored papules).

References

1. V. Marks et al. (2020). 'Chondrodermatitis Nodularis Helicis Differential diagnosis' Medscape. Available at: <https://emedicine.medscape.com/article/1119141-differential#1> accessed: 13 November 2023.

2. G. Gupta et al. (2023). 'Chondrodermatitis Nodularis Helicis'. National Library of Medicine. Available at: <https://www.ncbi.nlm.nih.gov/books/NBK482507/> accessed: 13 November 2023.

158 words

Reply



Re: Day 2-3 Scenario 2

by [Derek Kayanja \(Tutor\)](#) - Wednesday, 15 November 2023, 12:02 PM

Oyeombiliyetu, thank you for your useful addition to our discussion

10 words

Reply



Re: Day 2-3 Scenario 2

by [Derek Kayanja \(Tutor\)](#) - Wednesday, 15 November 2023, 12:01 PM

Thank you Rahima, good post.

5 words

Reply



Re: Day 2-3 Scenario 2

by [Felica Mwinji Namukonde](#) - Wednesday, 15 November 2023, 12:35 PM

Hey Rhima

thank you for your question DDX i came across include;

actinic keratosis

basal cell carcinoma

keratoacanthoma

squamous cell carcinoma and tophus

Reference

Goldstein A O Uptodate Available at: https://www.uptodate.com/contents/overview-of-benign-lesions-of-the-skin?search=chondrodermatitis%20nodularis%20helicis&source=search_result&selectedTitle=1~2&usage_type=
Accessed (15/11/23)

33 words

Reply



Re: Day 2-3 Scenario 2

by [Mostafa Damghani](#) - Tuesday, 14 November 2023, 10:19 PM

The Diagnosis for this scenario is likely to be Chondrodermatitis Nodularis Helix which is a painful, inflammatory benign condition that affects the cartilage and the skin of the outer ear. The diagnosis can be made based on previous history of pressure on the ear, predominante side when sleeping, extended use of headphones and etc. the nodules look scaly and crusted with surrounding erythema.

As Adam has already mentioned as well, dermoscopic features of CNH can help to differentiate between squamous cell carcinoma. dermoscopic features are lack of pigmentation, with radial white lines that end up to a yellow -brown centre that is covered by keratin.

Treatmen options are:

1st line: Relieve pressure on the lesion: using hollowed-out foam or pillows.

2nd line: steroid injection into the lesion to help with the inflammation and pain
cryotherapy and/or Use of 2% nitoglycerin BD to help with the prssure in arterial smooth muscle.

3rd line: Surgery: removing the abnormal cartilage.

dermnetnz.org. (n.d.). *Chondrodermatitis nodularis helicis (CNH)* | *DermNet NZ*. [online] Available at: <https://dermnetnz.org/topics/chondrodermatitis-nodularis>.

Griffiths, C., Barker, J., Bleiker, T., Chalmers, R. and Creamer, D. (2016). *Rook's Textbook of Dermatology, 4 Volume Set*. Wiley 108.9 Chondrodermatitis nodularis.



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26%

194 words

Reply



Re: Day 2-3 Scenario 2

by [Derek Kayanja \(Tutor\)](#) - Wednesday, 15 November 2023, 12:02 PM

Thank you Mostafa, good post.

5 words

Reply



Re: Day 2-3 Scenario 2

by [Felica Mwinji Namukonde](#) - Wednesday, 15 November 2023, 12:19 PM

Question1

Chondrodermatitis Nodularis Helicis is the diagnosis usually occurs as a solitary nodule less than 10mm in size located on the helix it mostly results from cartilage inflammation and that when its severe it may cause cartilage necrosis and transepithelial elimination of degenerated collagen.



Causes Include;

chronic trauma, chronic sun exposure, lower temperatures and prolonged pressure which compromises blood supply.

Associated systemic disease include;

connective tissue disease and cardiovascular disease, in this case our patient has a history of CVA which supports our diagnosis.

it is common in middle aged or older male patients.

the surface is usually covered with scale or crust that conceals a small ulcer. painful when pressure is applied

Diagnosis is based on clinical appearance and a positive history of rapid development of nodular lesion with pain and tenderness.

if unsure histopathology may help.

Treatment Include;

relieving pressure by use of donut-shaped pillow or foam padding for many weeks to relieve symptoms.

Intralesional triamcinolone injection which may provide long term resolution sometimes monthly treatments may help 3-4 sessions.

Surgical include; cryotherapy

curettage carbon dioxide laser ablation, wedge excision and cartilage removal.

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[search=chondrodermatitis%20nodularis%20helicis&source=search_result&selectedTitle=1~2&usage_type=defa](https://www.uptodate.com/contents/overview-of-benign-lesions-of-the-skin?search=chondrodermatitis%20nodularis%20helicis&source=search_result&selectedTitle=1~2&usage_type=defa) Accessed; (15/11/2023)

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205 words

[Reply](#)**Re: Day 2-3 Scenario 2**by [Derek Kayanja \(Tutor\)](#) - Friday, 17 November 2023, 11:24 AM

Thank you Felica, good post.

5 words

[Reply](#)**Re: Day 2-3 Scenario 2**by [Sandeep Kumar](#) - Wednesday, 15 November 2023, 1:34 PM

Thank you.

A good clinical case very commonly seen in the GP surgeries.

Although not routinely needed for this type of lesion, I would rely on dermatoscopy as it will show the classical daisy pattern.

The provisional clinical diagnosis from history and the picture will be Chondrodermatitis nodularis helcis.

Given the history of CVA it is possible that the patient has preference to sleeping on the effected site and there is a pressure contact point of ear with the pillow. This is the most common cause of CDHN.

There are many ways this can be treated as mentioned by the other members on this group but in this case, I would be more inclined on conservative management and advise on change n sleeping position or if that is not possible then using a pillow with hole in it to accommodate the ear and prevent further pressure damage. This is likely to regress the lesion in a few weeks to months and eliminate the need of any further aggressive treatment.

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230 words

[Reply](#)**Re: Day 2-3 Scenario 2**by [Derek Kayanja \(Tutor\)](#) - Friday, 17 November 2023, 11:25 AM

Thank you Sandeep, good post.

5 words

[Reply](#)**Re: Day 2-3 Scenario 2**by [Ayten Sadek](#) - Thursday, 16 November 2023, 2:28 AM

1-The likely diagnosis is Chondrodermatitis nodularis helcis

Chondrodermatitis nodularis helcis is a benign painful inflammatory condition affecting the ear. It typically involves a tender nodule or lump on the helix. The lesion can bleed and produce discharge. The exact cause is unclear, but factors such as pressure, trauma, or poor blood supply to the cartilage may cause it and usually the patient sleeps on the affected side.

2-The bedside test that could be performed to assist the diagnosis is applying pressure to the nodule to see if it will reproduce pain.

3-Treatment options:

The management mainly depends on reducing pressure on the affected ear by sleeping on the other side, Using padded cushions or pillows to reduce friction and pressure, using foam 'ear protectors' may also help during sleep

Other treatment options:

Topical Treatments: Applying topical steroids to reduce inflammation and prevent infection.

Applying topical nitroglycerin to cause vasodilatation and relieve symptoms by improving perfusion to the affected area.

Implant: By using a filler to provide cushioning layer

Cryotherapy: Freezing the lesion with liquid nitrogen may be considered

Surgical Intervention: surgical excision might be recommended for persistent or severe cases

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217 words

Reply



Re: Day 2-3 Scenario 2

by [Jillian Simpson](#) - Thursday, 16 November 2023, 2:55 PM

Hi all, sorry I am late to the thread, some great posts here on chondrodermatitis nodularis helices. I have never seen this in practice in secondary care, so this is new learning for me.

It was interesting to read further that it is thought to result from cartilage inflammation that in extreme cases may cause cartilage necrosis and transepithelial elimination of degenerated collagen. (Goldstein, 2023) I would be keen to know if anyone has seen this happen in practice?

As Adam and others have mentioned it most often occurs in middle-aged or older fair skinned males but it can occur in females and younger patients too (Eshraghi, 2018). It typically develops on the same side a patient lies, it enlarges rapidly over a few months then remains stable. It will cause pain by pressure or by changes in ambient temperature which is why wearing warm hats can help with the conservative management of it.

Diagnosis is based upon the clinical appearance and history of rapid development of a nodular lesion with associated pain and tenderness. In men the most common site for chondrodermatitis nodularis helices is the helix whereas in woman it is more common on the antihelix (Eshraghi, 2018). If the diagnosis is in doubt, the lesion should be excised and sent for histopathologic examination. Histology shows a nodule of degenerated collagen surrounded by a lymphohistiocytic infiltrate; a central ulceration through which the degenerated collagen is extruded is often present (Goldstein, 2023).

From reading through treatment options it appears relieving pressure is the most important aspect of managing these lesions. The use of a donut-shaped pillow specifically for this condition for several weeks may help resolve symptoms (PCDS, 2022). If the hole is not in the most appropriate position then the patient may wish to make their own from cutting a hole into a foam pillow (PCDS, 2022).

When therapeutic treatments are needed then intralesional triamcinolone acetonide injection at a dose of 10 mg/mL may provide rapid pain relief and long-term resolution of chondrodermatitis nodularis helices in some but patients may need monthly treatments for 3 to 4 months (Goldstein, 2023).

Surgery by the way of curettage or excision can be curative but recurrence rates of up to 30% have been reported (PCDS, 2022).

References:

Goldstein, A. (2023) Overview of benign lesions of the skin. UpToDate. [Online] Available at:

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432 words

Reply



Re: Day 2-3 Scenario 2

by [Derek Kayanja \(Tutor\)](#) - Friday, 17 November 2023, 11:26 AM

Jillian, thank you good contribution.

5 words

Reply



Re: Day 2-3 Scenario 2

by [Derek Kayanja \(Tutor\)](#) - Friday, 17 November 2023, 11:25 AM

Thank you Ayten, good post.

5 words

Reply



Re: Day 2-3 Scenario 2

by [Derek Kayanja \(Tutor\)](#) - Friday, 17 November 2023, 11:31 AM

Hi Group,

Thank you all who contributed to this discussion. This is as you have all correctly stated a case of Chondrodermatitis nodularis helcis (CHN), the bedside test is to squeeze the nodule with light pressure to demonstrate the conditions key feature which is its tenderness, hence light squeezing promoting pain is virtually pathognomonic. The most appropriate treatment for this patient is Conservative management with an ear pressure saving device. CHN is a benign inflammatory response to sustained pressure on the skin. In this case, the history of stroke is highly relevant.

93 words

Reply



Re: Day 2-3 Scenario 2

by [Siddharth Saluja](#) - Saturday, 18 November 2023, 11:09 AM

1) To assist in the diagnosis of Chondrodermatitis nodularis helcis (CNH), a thorough clinical examination is usually sufficient. However, a bedside test called the "top hat sign" can be performed. This involves applying pressure around the lesion with a cylindrical object, such as a pen or a cotton swab. If the patient experiences pain or tenderness upon compression, it supports the diagnosis of CNH.(1)

2) The likely diagnosis based on the provided information is Chondrodermatitis nodularis helcis. CNH is a benign inflammatory condition that typically affects the helix or antihelix of the ear. It presents as a small, painful, well-defined nodule, often with a central crust or erosion.(1)

3) The most appropriate treatment for CNH involves a combination of conservative measures and, in some cases, surgical intervention. Conservative treatment options include avoiding pressure on the affected area, using protective padding, applying topical antibiotics or corticosteroids, and maintaining good hygiene. In cases where conservative measures fail, surgical excision, cryotherapy, or laser therapy may be considered.(2)

References

1)dermnetnz.org. (n.d.). Chondrodermatitis nodularis helcis (CNH) | DermNet NZ. [online] Available at: <https://dermnetnz.org/topics/chondrodermatitis-nodularis>.

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192 words

Reply



Re: Day 2-3 Scenario 2

by [Derek Kayanja \(Tutor\)](#) - Sunday, 19 November 2023, 9:02 AM

Thank you Siddharth, when conducting bedside tests light squeezing with ones fingers promoting pain is virtually pathognomonic of CNH.

19 words

Reply



Re: Day 2-3 Scenario 2

by [Sadaf Basharat](#) - Sunday, 19 November 2023, 5:22 PM

1)Chondrodermatitis nodularis helicis is the most likely diagnosis for the described lesion.

It is a benign condition characterized by solitary, painful nodules commonly found on the ear helix or anti helix. Often called as wrinkler's disease. The nodules are usually scaly and crusted with underlying ulceration, and are surrounded by erythema. (Eshraghi and Oakley, 2018; Cunliffe, 2022).

Given the patient's history of right hemiparesis from a previous stroke, might had contributed to the development of this lesion. The usual causes of CNH include repeated trauma, exposure to cold, consistently sleeping on the same side, prolonged use of hearing aids and head phones.

Dermatoscopy, a diagnostic technique, can reveal a distinct "daisy pattern" when examining this lesion.

Treatment:

Treatment options include.

>. conservative

1. Use pressure-relieving prostheses/padding for cost-effective relief.
2. Topical and intralesional steroids have limited effectiveness without addressing pressure.
3. Hyaluronic acid injection offers cushioning and insulation.
4. Photodynamic therapy may be effective but requires multiple sessions.
5. Carbon dioxide/argon laser treatment involves lesion ablation for healing.
6. Topically apply 2% nitroglycerine for improved perfusion.
7. Diltiazem cream is considered an effective alternative.
8. Cryosurgery is an option for lesion removal.
9. Electrocauterization/curettage is an alternative to surgical excision.

>Surgical:

Various techniques for CNH excision include wide local excision with auricular reconstruction, lesion skin, and cartilage removal. Primary excision with meticulous cartilage trimming is preferred for satisfactory results.

References:

- 1.Nielsen, L.J., Olsen, C.H. and Lock-Andersen, J., 2016. Therapeutic options of chondrodermatitis nodularis helicis. *Plastic Surgery International*, 2016.
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276 words

Reply



Re: Day 2-3 Scenario 2

by [Bhavika Patel](#) - Tuesday, 21 November 2023, 5:50 PM

Chondrodermatitis nodularis helcis (CNH) is the likely diagnosis given that the tender nodular lesion is on a helical pressure point and has the characteristic erythematous scaly appearance. Given this patient has had a right sided hemiparesis, they are likely to be laying on the right side when sleeping. The image appears to be from a right ear given the anatomical orientation.

Chondrodermatitis nodularis chronica helcis (CNCH)

Also known as Wrinkler's disease, named after a dermatologist called Max Wrinkler in 1915 who established the 'Wrinkler's nodule' (Eshraghi, 2018). Commonly presents as a benign tender erythematous nodule fixed to the auricular helix or antihelix (Salah, 2018). 'Chondrodermatitis' meaning cartilaginous inflammatory reaction, 'nodularis' describing the nodular appearance, 'chronica' denoting to the chronic history to develop, and 'helicis' to describe the location affected as such the auricular helix.

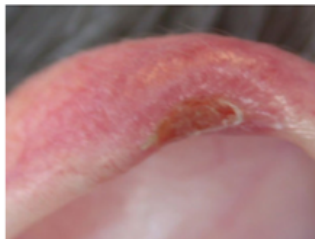
The demographics of the most affected populations include males aged 40 to 80 years with a male to female ratio of 10:1. One theory is that women tend to have more hair to protect the helix and instead develop chondrodermatitis of the antihelix instead.

Typically unilateral lesions appear but bilateral lesions are known for patients switching sides they sleep on. A bedside test would be the 'pressure test'. Applying gentle pressure to the nodule will elicit pain which would be consistent with CNCH rather than a BCC or SCC.

Chondrodermatitis nodularis



Chondrodermatitis



Chondrodermatitis



Image provided by Dr Trevor Evans

The etiopathogenesis of CNCH (Salah, 2018):

- Repetitive microtrauma affects vascular supply (recurrent headset use, piercings, hats or ear devices)
- Pressure from sleeping on the same side interferes with blood supply therefore causing more damage to the cartilage for the pressurised area.
- Increasing age causing thinning of the cartilage and skin, reduced elasticity, and loss of vascular or cartilaginous structures. Elderly people often have minimal movement when sleeping therefore causing prolonged excessive pressure at the same pressure point.
- Pressure necrosis from degeneration of cartilage and arteriolar narrowing contributing to perichondrial vasculitis theory (ischaemic death of cartilage).
- Anatomically poor vascular supply of the helix contributes to ischaemia.
- Possible genetic factor involved.

- Growth of "glomus-like vascular changes" causing glomus tumours.
- Cartilaginous disorders can exacerbate CNCH due to damage to dermal cartilage.
- Autoimmune conditions such as thyroiditis, SLE or scleroderma can cause younger patients to be affected by CNCH.

Management of CNCH (Salah, 2018):

Conservative measures are preferred to allow spontaneous resolution without invasive measures. However, in cases of atypical presentations, a biopsy may be needed to exclude a malignant change or cause.

Conservative measures include supported pillows and pressure-relieving devices, sleeping on alternate sides and applying regular ointments as a preventative measure.

Intralesional steroid injections include triamcinolone acetonide which has a "success rate of 33%". Recurrent injections are usually needed.

Topical nitroglycerine can cause vasodilation to theoretically reverse pressure necrosis to the cartilage. I have not come across this use in primary care as I wouldn't be aware on how to monitor the use.

Photodynamic therapy could be used to improve the vascular supply however there is not much evidence for this use.

Surgical wedge excision of the affected area comes with post-procedural complications such as infection and scarring. This can be reserved for serious cases.

Curettage can be used to remove the necrotic tissue to prevent further damage however there is a recurrence rate.

Eshraghi, A (2018). Chondrodermatitis nodularis. DermNet NZ. [Online]. Available at: <https://dermnetnz.org/topics/chondrodermatitis-nodularis> [Accessed 19th November 2023]

Salah, H., Urso, B., Khachemoune, A. and Urso, B.A., 2018. Review of the etiopathogenesis and management options of chondrodermatitis nodularis chronica helcis. *Cureus*, 10(3).

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600 words

Reply



Re: Day 2-3 Scenario 2

by [Arwaf Benrawwaf](#) - Monday, 11 December 2023, 3:42 PM

1. A dermatoscopy or skin surface microscopy could be performed to assist in the diagnosis of the nodule on the patient's pinna. Dermatoscopy allows for a closer examination of skin lesions and can help differentiate between benign and malignant lesions (Lallas et al., 2013)(Morgado-Carrasco,2019)

2. Chondrodermatitis Nodularis Helicis (CNH) is a common benign condition that presents as a tender nodule on the helix or antihelix of the ear, typically in middle-aged or older individuals. The etiology is thought to be due to repetitive mechanical trauma or pressure, leading to ischemia and subsequent inflammation and necrosis of the cartilage and overlying skin. The lesion is typically well-circumscribed, dome-shaped, and covered by a central crust or scale. Unlike the patient's presentation, CNH is not usually associated with bleeding or ulceration (dermnetnz.org.2018)



3. The most appropriate treatment for this patient would be surgical excision of the lesion with a clear margin. Depending on the size and location of the lesion, Mohs micrographic surgery may be considered to ensure complete removal of the cancerous tissue (Schmitt et al., 2018). Additionally, the patient may benefit from radiation therapy to reduce the risk of recurrence (National Comprehensive Cancer Network, 2022). As the patient is already on aspirin, it may be continued as it has been shown to reduce the risk of metastasis in some types of cancer. However, further discussion with the patient's healthcare provider is necessary to determine the most appropriate treatment plan for the patient.

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369 words

Reply

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Scenario 3 ▶

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